

Assignment 8 Solutions

9.1 $n = 6$, $\text{Nul } A = \{ \vec{v} : \underset{\substack{\uparrow \\ (m \times n) \\ \downarrow \\ (n \times 1) \\ (3 \times 6)}}{A} \vec{v} = \vec{0} \}$

9.2 $\dim \text{Nul } A \geq 3$, $\text{rank } A + \dim \text{Nul } A = 13$

$$\text{rank } A = \dim \text{Col } A \leq 10$$

$$\dim \text{Nul } A = 13 - \text{rank } A \geq 3$$

9.3 $\dim \text{Nul } A = 1$ $\dim \text{Nul } A = 5 - \text{rank } A = 5 - 4 = 1$

9.8 $\text{Col } A = \text{span} \{ \vec{0} \} = \text{span} \{ \}$

$$\text{Nul } A = \mathbb{R}^2 = \text{span} \{ \vec{e}_1, \vec{e}_2 \} = \text{span} \left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\}$$

9.11 $\begin{bmatrix} 1 & -4 & -3 \\ -3 & 12 & 10 \\ -2 & 8 & 8 \\ -1 & 4 & 2 \end{bmatrix} \sim \begin{bmatrix} 1 & -4 & -3 \\ 0 & 0 & 1 \\ 0 & 0 & 2 \\ 0 & 0 & -1 \end{bmatrix} \sim \begin{bmatrix} 1 & -4 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

$$\text{Col } A = \text{span} \left\{ \begin{bmatrix} 1 \\ -3 \\ -2 \\ -1 \end{bmatrix}, \begin{bmatrix} -3 \\ 10 \\ 8 \\ 2 \end{bmatrix} \right\}$$

$$\begin{aligned} x_1 &= 4x_2 \\ x_2 &\text{ is free} \\ x_3 &= 0 \end{aligned} \quad x_2 \begin{bmatrix} 4 \\ 1 \\ 0 \end{bmatrix}$$

$$\text{Nul } A = \text{span} \left\{ \begin{bmatrix} 4 \\ 1 \\ 0 \end{bmatrix} \right\}$$

9.12

$$\text{Col } A = \text{span} \{ \vec{a}_1, \vec{a}_2, \vec{a}_4, \vec{a}_5, \vec{a}_7 \}$$

$$x_1 = -3x_3 - 5x_6 + 7x_8$$

$$x_2 = -14x_3 - 2x_6 - x_8$$

 x_3 is free

$$x_4 = -3x_6 - 7x_8$$

$$x_5 = -8x_6 - 9x_8$$

 x_6 is free

$$x_7 = -x_8$$

 x_8 is free

$$x_3 \begin{bmatrix} -3 \\ -14 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} + x_6 \begin{bmatrix} -5 \\ -2 \\ 0 \\ -3 \\ -8 \\ 1 \\ 0 \\ 0 \end{bmatrix} + x_8 \begin{bmatrix} 7 \\ -1 \\ 0 \\ -7 \\ -9 \\ 0 \\ -1 \\ 1 \end{bmatrix}$$

$$\text{Nul } A = \text{span} \left\{ \begin{bmatrix} -3 \\ -14 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} -5 \\ -2 \\ 0 \\ -3 \\ -8 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 7 \\ -1 \\ 0 \\ -7 \\ -9 \\ 0 \\ -1 \\ 1 \end{bmatrix} \right\}$$

$$9.15 \quad \begin{bmatrix} 1 & -1 & 2 & -2 \\ -1 & 0 & -4 & -4 \\ 0 & -3 & 18 & 6 \end{bmatrix} \sim \begin{bmatrix} 1 & -1 & 2 & -2 \\ 0 & 1 & -6 & -2 \\ 0 & -3 & 18 & 6 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & -4 & -4 \\ 0 & 1 & -6 & -2 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad \text{Col } A = \text{span} \left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ -3 \end{bmatrix} \right\}$$

$$9.17 \quad \begin{bmatrix} 1 & -1 & -2 & 4 \\ -1 & 2 & 2 & 1 \\ 0 & 1 & 1 & 3 \end{bmatrix} \sim \begin{bmatrix} 1 & -1 & -2 & 4 \\ 0 & 1 & 0 & 5 \\ 0 & 1 & 1 & 3 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & -2 & 9 \\ 0 & 1 & 0 & 5 \\ 0 & 0 & 1 & -2 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 & 5 \\ 0 & 1 & 0 & 5 \\ 0 & 0 & 1 & -2 \end{bmatrix}$$

$$[\vec{u}]_{\mathcal{B}} = \begin{bmatrix} 5 \\ 5 \\ -2 \end{bmatrix}$$

9.24 False $\mathbb{R}^2 = \text{span} \left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\} = \text{span} \left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} -1 \\ 2 \end{bmatrix} \right\}$

9.25 False

$$A = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad \dim \text{Nul } A = 0$$

9.28 True $\vec{b} = [\vec{a}_1, \dots, \vec{a}_n] \begin{bmatrix} v_1 \\ \vdots \\ v_n \end{bmatrix} = \sum_{i=1}^n v_i \vec{a}_i \in \text{span} \{ \vec{a}_i \}_{i=1}^n = \text{Col } A$

9.31 False $\vec{x} \mapsto \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \vec{x} \quad T(\vec{e}_3) = \vec{0}$

9.32 True $\begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} = \sum_{i=1}^n x_i \vec{e}_i \Rightarrow [\vec{x}]_{\mathcal{B}}$

9.33 False $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$

9.36 $\{ \vec{v}_1, \vec{v}_3 \}, \{ \vec{v}_1, \vec{v}_4 \}, \{ \vec{v}_2, \vec{v}_3 \}, \{ \vec{v}_2, \vec{v}_4 \}, \{ \vec{v}_3, \vec{v}_4 \}$

(a)

$$9.39 \quad \begin{bmatrix} 1 & 2 & 3 & -1 \\ 1 & 2 & 4 & -4 \\ 0 & 0 & 2 & 4 \\ 0 & 0 & 1 & 2 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & 3 & -1 \\ 0 & 0 & 1 & -3 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & 3 & -1 \\ 0 & 0 & 1 & -3 \\ 0 & 0 & 0 & 5 \\ 0 & 0 & 0 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$x_1 = -2x_2$$

$$x_2 \text{ is free}$$

$$x_3 = 0$$

$$x_4 = 0$$

$$\text{Nul } A = \text{span} \left\{ \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix} \right\}$$

(b)

$$\begin{bmatrix} 1 & 2 & 3 & -1 & b_1 \\ 1 & 2 & 4 & -4 & b_2 \\ 0 & 0 & 2 & 4 & b_3 \\ 0 & 0 & 1 & 2 & b_4 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & 3 & -1 & b_1 \\ 1 & 2 & 4 & -4 & b_2 \\ 0 & 0 & 1 & 2 & b_4 \\ 0 & 0 & 0 & 0 & b_3 - 2b_4 \end{bmatrix}$$

$$x_3 - 2b_4 = 0$$

9.41

$$R_2 \leftarrow R_2 + 2R_1$$

$$R_3 \leftarrow R_3 - 7R_1$$

$$R_3 \leftarrow R_3 - 5R_2$$

$$R_4 \leftarrow R_4 - R_3$$

$$A \longrightarrow U$$

$$\begin{bmatrix} b_1 \\ b_2 \\ b_3 \\ b_4 \end{bmatrix} \xrightarrow{R_2 \leftarrow R_2 + 2R_1} \begin{bmatrix} b_1 \\ b_2 + 2b_1 \\ b_3 \\ b_4 \end{bmatrix} \xrightarrow{R_3 \leftarrow R_3 - 7R_1} \begin{bmatrix} b_1 \\ b_2 + 2b_1 \\ b_3 - 7b_1 \\ b_4 \end{bmatrix} \xrightarrow{R_3 \leftarrow R_3 - 5R_2} \begin{bmatrix} b_1 \\ b_2 + 2b_1 \\ b_3 - 7b_1 - 5(b_2 + 2b_1) \\ b_4 \end{bmatrix}$$

$$\begin{bmatrix} b_1 \\ b_2 + 2b_1 \\ b_3 - 7b_1 - 5(b_2 + 2b_1) \\ b_4 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 + 2b_1 \\ -17b_1 - 5b_2 + b_3 \\ b_4 \end{bmatrix} \xrightarrow{R_4 \leftarrow R_4 - R_3} \begin{bmatrix} * \\ * \\ * \\ b_4 - (-17b_1 - 5b_2 + b_3) \end{bmatrix}$$

$$b_4 + 17b_1 + 5b_2 - b_3$$

$$17x_1 + 5x_2 - x_3 + x_4 = 0$$

9.42 (a) Closure under addition:

$$\vec{h}_1 \in H \Rightarrow \vec{h}_1 = \vec{u}_1 + \vec{v}_1 \text{ where } \vec{u}_1 \in U, \vec{v}_1 \in V$$

$$\vec{h}_2 \in H \Rightarrow \vec{h}_2 = \vec{u}_2 + \vec{v}_2 \text{ where } \vec{u}_2 \in U, \vec{v}_2 \in V$$

$$\vec{h}_1 + \vec{h}_2 = (\vec{u}_1 + \vec{v}_1) + (\vec{u}_2 + \vec{v}_2) = (\vec{u}_1 + \vec{u}_2) + (\vec{v}_1 + \vec{v}_2)$$

$$\vec{u}_1 + \vec{u}_2 \in U \text{ since } U \text{ is CUA}$$

$$\vec{v}_1 + \vec{v}_2 \in V \text{ since } V \text{ is CUA}$$

$$\text{So } \vec{h}_1 + \vec{h}_2 \in H \quad \checkmark$$

Closure under scaling:

$$\vec{h} \in H \Rightarrow \vec{h} = \vec{u} + \vec{v} \text{ where } \vec{u} \in U, \vec{v} \in V$$

$$c\vec{h} = c(\vec{u} + \vec{v}) = c\vec{u} + c\vec{v}$$

$$c\vec{u} \in U \text{ since } U \text{ is CUS}$$

$$c\vec{v} \in V \text{ since } V \text{ is CUS}$$

$$\text{So } c\vec{h} \in H \quad \checkmark$$

$$(b) \begin{bmatrix} 6 & -6 & -1 & 17 & -8 & 5 & -16 & 1 \\ 9 & -2 & 2 & 36 & 2 & 11 & 4 & 16 \\ -10 & -13 & 8 & -45 & 3 & -2 & 6 & 3 \\ -9 & 4 & -1 & -33 & 2 & -10 & 4 & 0 \\ 2 & 4 & 2 & 14 & 8 & 4 & 16 & 9 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 & 4 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 1 & 0 & 2 & 0 \\ 0 & 0 & 1 & 1 & 2 & 1 & 4 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\dim H = 4$$